Zoho DSA Prep Roadmap

Expanded Edition | Prepared for Mohan

Start date: 28 July 2026 | 175 problems across 15 topics | 6-month plan

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Reality check before you start

This is your third study resource this cycle. The Zoho roadmap, the aptitude handbook, and now this — all well-built, none of them worth anything if they sit unopened after week one. That has been the pattern so far: build the resource, feel productive, don't touch it daily. The aptitude quizzes only moved your score (47% to 13/15) because you showed up daily, not because the material was good. This document is not the work. Opening your editor and solving problem 1 today is the work. Track it daily — a simple checklist, a streak count, anything — or this becomes another PDF you show people instead of a plan you followed.

What changed from your original roadmap: nothing was removed. 50 problems were added across the existing 14 topics plus one new topic (Hashing) that Zoho Round 2 tests directly but your original list didn't cover on its own. Every addition is marked **[NEW]**. Total load only moved from ~0.8 problems/day to ~1.15 problems/day against your stated 3-4/day capacity — the extra problems do not break your timeline. What breaks your timeline is not solving daily.

Timeline at a Glance

Month	Dates	Topics	Problems	Watch for
1	28 Jul - 27 Aug '26	Arrays, Sorting	34	Build the daily habit first
2	28 Aug - 27 Sep '26	Binary Search, Strings	30	Strings = heaviest Zoho topic
3	28 Sep - 27 Oct '26	Sliding Window/2-Pointer, Linked List	26	-
4	28 Oct - 27 Nov '26	Recursion & Backtracking, Bit Manip., Stack & Queue	31	Check your Sem 5 exam dates now
5	28 Nov - 27 Dec '26	Trees, BST, Pattern Printing, Hashing (+Graphs/DP optional)	54*	Likely clashes with end-sem exams
6	28 Dec '26 - 27 Jan '27	Revision + timed mocks only	-	No new topics

**Month 5 is deliberately heavy. Graphs and DP are the original roadmap's optional safety net — if your semester exams eat into this month, drop them first. Trees, BST, Pattern Printing, and Hashing are not optional; they are Zoho's actual Round 2 territory.*

Revision structure for Month 6: Week 1-2 re-solve every problem you marked difficult on the first pass (don't re-solve ones you already found easy — that's comfort, not revision). Week 3 full timed mocks, one per topic cluster. Week 4 timed mixed mocks simulating the actual Round 2 format.

MONTH 1 (28 Jul - 27 Aug 2026) - Arrays, Sorting

Topic 1: Arrays (24 problems)

Your highest-weight topic by far.

1. Find the Largest Element in an Array
2. Find the Second Largest Element (without sorting)
3. Leaders in an Array (element greater than everything to its right)
4. Reverse an Array In-Place (two-pointer swap)
5. Rotate an Array by K Positions
6. Move All Zeroes to the End
7. Find the Missing Number (1 to N)
8. Find Duplicate(s) in an Array
9. Equilibrium Index of an Array
10. Union and Intersection of Two Sorted Arrays
11. Kadane's Algorithm (Maximum Subarray Sum)
12. Maximum Product Subarray
13. Best Time to Buy and Sell Stock (single transaction)
14. Rearrange Array in Alternating Positive/Negative Order
15. Next Permutation of an Array
16. Print All Permutations of an Array of Numbers
17. Pascal's Triangle (generate a row / the full triangle)
18. Merge Overlapping Intervals
19. Trapping Rain Water **[NEW]**
20. Merge Two Sorted Arrays Without Extra Space **[NEW]**
21. Majority Element (Moore's Voting Algorithm) **[NEW]**
22. Find All Pairs with a Given Difference **[NEW]**
23. Rearrange Array Such That arr[i] = i **[NEW]**
24. Sort an Array by Parity (Even Before Odd) **[NEW]**

Topic 2: Sorting Techniques (10 problems)

TCS and Zoho both occasionally ask you to write a sort from scratch, not just call a library function.

1. Implement Bubble Sort from scratch
2. Implement Selection Sort from scratch
3. Implement Insertion Sort from scratch
4. Implement Merge Sort (Divide & Conquer)
5. Implement Quick Sort (partition logic)
6. Count Inversions in an Array (using Merge Sort)
7. Sort an Array of 0s, 1s, and 2s (Dutch National Flag)
8. Cyclic Sort (for missing/duplicate number problems) **[NEW]**
9. Heap Sort from Scratch **[NEW]**
10. Sort a Nearly Sorted (K-Sorted) Array **[NEW]**

MONTH 2 (28 Aug - 27 Sep 2026) - Binary Search, Strings

Topic 3: Binary Search **(11 problems)**

Kept matrix-light on purpose.

1. Binary Search on a Sorted Array (iterative + recursive)
2. First and Last Occurrence of an Element
3. Count Occurrences of an Element in a Sorted Array
4. Search in a Rotated Sorted Array
5. Find the Minimum Element in a Rotated Sorted Array
6. Find the Peak Element
7. Square Root of a Number using Binary Search
8. Search Insert Position (lower bound)
9. Median of Two Sorted Arrays **[NEW]**
10. Kth Smallest Element Using Binary Search on Answer **[NEW]**
11. Allocate Minimum Number of Pages/Books (Binary Search on Answer) **[NEW]**

Topic 4: Strings **(19 problems)**

The most heavily tested topic at Zoho.

1. Reverse a String In-Place (no built-in reverse)
2. Check if a String is a Palindrome
3. Toggle Case of Each Character in a String
4. Check if Two Strings are Anagrams
5. Check if a String is a Permutation of Another String
6. Count Frequency of Each Character in a String
7. Find the First Non-Repeating Character in a String
8. Remove Duplicate Characters from a String
9. String Compression / Run-Length Encoding & Decoding
10. Check if One String is a Rotation of Another
11. Check if a String Contains a Permutation of Another String as a Substring
12. Longest Palindromic Substring (expand-around-center)
13. Reverse Words in a Sentence (in-place, no split())
14. Implement atoi() (String to Integer, no library conversion)
15. Longest Common Prefix of an Array of Strings
16. Minimum Window Substring **[NEW]**
17. Check if a String is a Valid Shuffle of Two Other Strings **[NEW]**
18. KMP Pattern Matching (basic implementation) **[NEW]**
19. Group Anagrams from a List of Strings **[NEW]**

MONTH 3 (28 Sep - 27 Oct 2026) - Sliding Window / Two Pointer, Linked List

Topic 5: Sliding Window / Two Pointer (11 problems)

A compact pattern set that shows up disguised as many different array/string problems.

1. Two Sum on a Sorted Array (two-pointer)
2. Pair with Given Sum in an Unsorted Array (hashing)
3. Maximum Sum Subarray of Size K (fixed window)
4. Longest Substring Without Repeating Characters
5. Smallest Subarray with Sum >= Given Value
6. Longest Substring with At Most K Distinct Characters
7. Triplet Sum to a Given Value (3Sum)
8. Container With Most Water
9. Longest Repeating Character Replacement **[NEW]**
10. Subarray with Given XOR **[NEW]**
11. Count Subarrays with At Most K Odd Numbers **[NEW]**

Topic 6: Linked List (15 problems)

Reverse/cycle-detection variants are asked almost verbatim in real Zoho and Cognizant drives.

1. Traverse a Linked List and Find its Length
2. Insert/Delete a Node at a Given Position
3. Reverse a Linked List (iterative)
4. Reverse a Linked List (recursive)
5. Find the Middle of a Linked List (slow/fast pointer)
6. Detect a Cycle in a Linked List (Floyd's Algorithm)
7. Find the Starting Node of the Loop
8. Merge Two Sorted Linked Lists
9. Remove the N-th Node From the End of the List
10. Check if a Linked List is a Palindrome
11. Add Two Numbers Represented as Linked Lists
12. Intersection Point of Two Linked Lists **[NEW]**
13. Clone a Linked List with Random Pointers **[NEW]**
14. Sort a Linked List (Merge Sort on Linked List) **[NEW]**
15. Flatten a Multilevel Linked List **[NEW]**

MONTH 4 (28 Oct - 27 Nov 2026) - Recursion & Backtracking, Bit Manipulation, Stack & Queue

Before this month starts

This window likely overlaps your Semester 5 end-semester exam period. Confirm your actual exam dates now, not in October. If they land inside this month, exams win — shift DSA to lighter maintenance (1 problem/day, just enough to not lose momentum) and pick this pace back up once exams are done. Don't try to run both at full intensity; that's how both get done badly.

Topic 7: Recursion & Backtracking (11 problems)

1. Factorial and Power of a Number (recursive)
2. Fibonacci Number (plain recursion)
3. Sum of Digits and Reverse a Number (recursive)
4. Check if a String/Number is a Palindrome (recursive)
5. Print All Subsequences/Subsets of an Array
6. Generate All Permutations of a String/Array
7. Generate Permutations of a String/Array with Duplicate Elements
8. N-Queens (basic - print one valid placement)
9. Rat in a Maze (Path Existence) **[NEW]**
10. Combination Sum **[NEW]**
11. Print the Power Set of a String (Recursive) **[NEW]**

Topic 8: Bit Manipulation (8 problems)

1. Check if a Number is Even or Odd using Bitwise AND
2. Count the Number of Set Bits in an Integer
3. Check if a Number is a Power of Two
4. Find the Single Non-Repeating Number in an Array (XOR)
5. Swap Two Numbers Without a Temporary Variable (XOR swap)
6. Find the Two Non-Repeating Elements in an Array (XOR-based)
7. Count Total Set Bits from 1 to N **[NEW]**
8. Generate Power Set Using Bitmasking **[NEW]**

Topic 9: Stack & Queue (12 problems)

Zoho is the company most likely to ask you to build these structures, not just use them.

1. Implement a Stack using an Array
2. Implement a Queue using an Array (circular queue)
3. Implement a Stack using a Queue (and vice versa)
4. Valid Parentheses (balanced brackets)
5. Min Stack (retrieve the minimum in O(1))
6. Next Greater Element
7. Evaluate a Postfix Expression
8. Convert Infix to Postfix
9. The Celebrity Problem (stack-based elimination)
10. Sliding Window Maximum (Deque-Based) **[NEW]**
11. Largest Rectangle in Histogram **[NEW]**
12. Implement an LRU Cache (Queue + HashMap) **[NEW]**

MONTH 5 (28 Nov - 27 Dec 2026) - Trees, BST, Pattern Printing, Hashing (+Graphs/DP optional)

Topic 10: Trees **(14 problems)**

Zoho's clearest differentiator from TCS/CTS - budget real time here if Zoho is the priority.

1. Inorder, Preorder, and Postorder Traversal (recursive)
2. Level Order Traversal (BFS with a Queue)
3. Height / Maximum Depth of a Binary Tree
4. Diameter of a Binary Tree
5. Check if Two Binary Trees are Identical
6. Check if a Binary Tree is Height-Balanced
7. Check if a Binary Tree is Symmetric (Mirror Image)
8. Root-to-Leaf Path Sum (does any path equal a target?)
9. Lowest Common Ancestor in a Binary Tree
10. Zig-Zag (Spiral) Level Order Traversal
11. Iterative Inorder/Preorder Traversal using a Stack
12. Construct Binary Tree from Inorder and Preorder Traversal **[NEW]**
13. Vertical Order Traversal of a Binary Tree **[NEW]**
14. Boundary Traversal of a Binary Tree **[NEW]**

Topic 11: Binary Search Trees **(9 problems)**

1. Search for a Value in a BST
2. Insert a Node into a BST
3. Delete a Node from a BST
4. Validate whether a Binary Tree is a Valid BST
5. Find the Floor and Ceil of a Value in a BST
6. Kth Smallest Element in a BST (via inorder traversal)
7. Lowest Common Ancestor in a BST
8. Convert a Sorted Array to a Balanced BST **[NEW]**
9. Inorder Successor of a Node in a BST **[NEW]**

Topic 12: Graphs - Optional Safety Net **(6 problems)**

Drop this first if Month 5 collides with exams.

1. Represent a Graph using an Adjacency List
2. BFS Traversal of a Graph
3. DFS Traversal of a Graph (recursive)
4. Detect a Cycle in an Undirected Graph **[NEW]**
5. Detect a Cycle in a Directed Graph **[NEW]**
6. Number of Islands (Grid BFS/DFS) **[NEW]**

Topic 13: Dynamic Programming - Optional Safety Net **(7 problems)**

Drop this first if Month 5 collides with exams.

1. Climbing Stairs / Fibonacci with Memoization
2. Maximum Subarray Sum, Revisited as DP (Kadane's)
3. House Robber (Maximum Sum, No Two Adjacent)
4. 0/1 Knapsack (basic 2D DP)
5. Longest Common Subsequence **[NEW]**
6. Longest Increasing Subsequence **[NEW]**
7. Coin Change (Minimum Number of Coins) **[NEW]**

Topic 14: Pattern Printing (Zoho Essentials) **(12 problems)**

Zoho heavily tests this in Round 2. Do these without looking up the answers.

1. Print a String in an 'X' Format (most frequently asked Zoho pattern question)
2. Print a Diamond Pattern (stars and incrementing numbers)
3. Print a Hollow Square / Rectangle Pattern (tests boundary conditions in loops)
4. Print a Number Pyramid / Floyd's Triangle
5. Print a Butterfly Pattern (mirrored loops)
6. Print a Spiral Matrix (numbers 1 to N^2 in a spiral)
7. Print a Snake Pattern Matrix (left-right row 1, right-left row 2)
8. Print Concentric Squares (center 1, surrounded by 2s, surrounded by 3s)
9. Look-and-Say Sequence Pattern (1, 11, 21, 1211, 111221...)
10. Print a Right-Angled Triangle with Continuously Increasing Numbers
11. Print a Staircase Pattern **[NEW]**
12. Print a Zig-Zag / Wave Number Pattern **[NEW]**

Topic 15: Hashing / HashMap-Based Problems **(6 problems)**

New topic - not in your original list, but this pattern shows up constantly in Zoho Round 2 that your original roadmap only touched in passing (inside Sliding Window). Worth its own focused pass.

1. Two Sum (Unsorted Array, using HashMap) **[NEW]**
2. Longest Consecutive Sequence **[NEW]**
3. Subarray Sum Equals K **[NEW]**
4. First Missing Positive **[NEW]**
5. Check for Duplicates Within K Distance in an Array **[NEW]**
6. Sort Characters in a String by Frequency **[NEW]**

MONTH 6 (28 Dec 2026 - 27 Jan 2027) - Revision + Timed Mock Tests

No new topics this month. This is where the previous 5 months either pay off or don't.

Week 1-2: Revisit every problem you flagged as difficult on first pass. Re-solve from scratch, not by re-reading your old solution.

Week 3: Full timed mocks, one per topic cluster (Arrays+Sorting, Strings+Searching, Trees+BST, Pattern Printing).

Week 4: Timed mixed mocks simulating the actual Zoho Round 2 format - multiple topics, fixed time limit, no notes.

Total problem count

125 original + 50 added (44 spread across your 14 topics, 6 in the new Hashing topic) = 175 problems, 6 months, averaging under 1.2/day against your own stated capacity of 3-4/day. The plan has slack built in on purpose. Use it to go deep on the topics you're weak in, not to skip days.